Note: Mining is the only method by which new bitcoins are created. The three above-mentioned methods simply help in circulating them.

What about securing this 'currency'?

"Hmm…so how secure is it? With other currencies, I've got banks and their robust systems protecting my money, but this sounds like it's all on me."

Well, that's both true and false. With bitcoins, you'll need to be a bit more careful but such is the case with banks too. If you're casual in handling your bank account numbers and passwords, there is a fairly high risk of you losing your money even from your supposedly 'safe' bank account.

Besides, the foundation of Bitcoin is laid using one of the principle elements of information security, which is cryptography. Unless somebody invents a supercomputer that can break the sophisticated algorithms at play in the Bitcoin network (some agencies are trying hard but even they haven't achieved any significant progress on this front yet), there's only a very small likelihood of the network getting breached.

This brings us to the second most common concern when dealing with currency—fraud. There have been very few successful attempts to fraud the Bitcoin network but even those didn't last long. Security researchers have tried to break in but they couldn't *(http://goo.gl/fSlsMZ).*

Still, let's assume that somebody does give it a shot. Since the sanctity of the Bitcoin network is dependent on the transaction block chain, the most common way to fraud the system would be to introduce malicious transactions in the chain. Carrying forward the earlier-mentioned example, let's assume that Andy has malicious intentions and he wants to cheat Mary. Right after his transaction with her, he creates another transaction in which he transfers the 10 BTC back to himself.

On an average, every 10 minutes a transaction block is added to the transaction block chain and once this is added, the miners stop working with the previous block and start on the one that's just added. At any given moment, there are thousands of miners putting together their processing power simultaneously to honestly mine bitcoins.

If Andy wants to introduce the malicious transaction in the chain, he has to create a new transaction block with that transaction in it, and challenge not only the transaction block in which his first transaction was but also all the ones added after that. The challenge for Andy is to solve the 'proof of work' puzzle for the block that has the genuine transaction and for each block in the chain, after that particular one. This boils down to the processing power available with him, competing with the processing power available with thousands of honest bitcoin miners. Unless Andy has resources to obtain that kind of processing power and a much bigger incentive than his investment, he's not going to make it.

The fraud that Andy was trying to commit in the above example is called double-spending, i.e., spending exactly the same money twice. In the banking system, banks and payment gateways usually monitor systems for double-spending. In the Bitcoin network, the network itself takes care of it, as long as the processing power of honest miners is more than that of a fraudster.

The Bitcoin network has another security enhancement that beats the current banking system, and that is anonymity. Similar to the banking system, in the Bitcoin network too your wallet is represented by a string of seemingly random characters but that's all they have in common on this front. The Bitcoin network doesn't need to know who you are, nor your address, age and other details—all of which are required by the banking system.

The pros and cons

"I've heard you out. But I am still sceptical…"

Like every other thing out there, bitcoins also ship with a list of pros and cons.

The pros:

- The Bitcoin network is decentralised, which means no single authority, bank or government can control or regulate it. Therefore, it is inflation-proof.
- There's no third party involved while transacting in bitcoins. It's all person-to-person.
- Even though the transactions are public, it provides complete anonymity. People can know how much balance a particular account has but they can't find out who owns that account.
- There are no bank fees, credit card charges, service charges, etc. A bitcoin can be transferred from one account to another at the click of a key, irrespective of the country, state or city.
- The transactions done in bitcoins are irreversible and thereby prevent merchants from fraud purchases.
- Anyone with an Internet connection can deal in bitcoins.

The cons:

- The network is highly volatile. Unless you're a miner, you'll have to buy bitcoins with alternate currency. The exchange rate of a bitcoin is purely governed by demand and supply. When it started out, the exchange rate was US$ .0008 per BTC and achieved a maximum rate of $1100 per BTC in November 2013 *(http://goo.gl/pqQ7L).*
- There is a potential for the bitcoin to lose value if a better cryptocurrency crops up.
- It's irrecoverable. Unfortunately, if you lose access to your bitcoin wallet, there's no way to get it back. You'll have to write it off.
- Governments are staying away from it.

What you might be wondering…

- ***Can the governments completely ban Bitcoin?***
  As mentioned earlier, bitcoins are not issued by any government, so they literally have no say in how it's circulated. This also means that they cannot completely